FUNCTIONS & GRAPHS 3

Exponential & Logarithmic Functions

CANONICAL PRACTICE BANK

10 CORE + 10 CHALLENGE + 10 ADVANCED • Full worked solutions

Practice rhythm: identify structure → choose method → solve → check restrictions → interpret.

CORE

C1 — Linear or exponential?
A: $12, 17, 22, 27, \ldots$ B: $12, 18, 27, 40.5, \ldots$
(a) Which is linear? (b) Which is exponential? (c) State the constant difference/ratio.

C2 — Build an exponential function
An exponential function has initial value 7. Whenever x increases by 1, the output is multiplied by 3. Write $f(x) = Ab^x$, find $f(2)$, and classify growth/decay.

C3 — Read the equation
For $f(x) = 5(3/2)^x$, determine the y-intercept, domain, range, horizontal asymptote, and whether the function increases or decreases.

C4 — Transformed exponential
For $f(x) = 2 \cdot 3^{x-1} - 6$, find the horizontal asymptote, range, $f(1)$ and the x-intercept.

C5 — Common base
Solve $8^{x-1} = 4^{x+1}$.

C6 — Exponential ↔ logarithmic form
Rewrite each in the other form: (a) $5^3 = 125$; (b) $\log_2(1/8) = -3$; (c) $7^0 = 1$.

C7 — Log laws
Evaluate exactly $\log_2 40 - \log_2 5 + \log_2 2$, showing the log-law step.

C8 — Inverse graph
The graph $y = 4^x$ contains $(-1, 1/4), (0, 1), (1, 4)$. State the inverse, the corresponding three inverse points, its domain/range, and its vertical asymptote.

C9 — Growth model
A population follows $P(t) = 800(1.05)^t$. State the initial population and growth rate, find $P(3)$, and interpret the factor 1.05.

C10 — Simple log equation
Solve $\log_3(x - 2) = 2$. State the domain restriction first.

CHALLENGE

H1 — Choose logarithms
Solve $5 \cdot 2^x = 17$. Give an exact logarithmic answer and a decimal answer to 3 s.f.

H2 — Quadratic in an exponential
Solve $9^x - 10(3^x) + 9 = 0$. State the substitution explicitly.

H3 — Domain-sensitive log equation
Solve $\log_2(x - 1) + \log_2(x - 3) = 3$. State the domain first and reject invalid roots.

H4 — Transformed logarithm
For $f(x) = 3\ln(2x - 4) - 1$, find the domain, vertical asymptote, $f(5/2)$ and the exact x-intercept.

H5 — Inverse of a transformed exponential
For $f(x) = 4e^{x/2} - 3$, state the domain/range, find $f^{-1}(x)$, then state the inverse domain/range.

H6 — Reverse discrete model (Phase 4 revised)
A quantity follows $P(t) = 320b^t$ and $P(3) = 625$. Find b, the percentage growth rate, and $P(5)$.

H7 — When do two models meet?
$A(t) = 300(1.04)^t$ and $B(t) = 250(1.08)^t$. Find when they are equal, in exact logarithmic form and numerically. Interpret the result.

H8 — Decay inequality
Solve $5(0.6)^x \leq 1$. Give an exact logarithmic boundary and a decimal approximation.

H9 — Reconstruct a transformed exponential
$f(x) = Ab^x + 1$, with $A > 0$, $b > 0$, $b \neq 1$, passes through $(0, 4)$ and $(2, 13)$. Find A, b, f, the horizontal asymptote and the range.

H10 — Change of base
Solve $\log_2 x + \log_4 x = 6$ exactly.

ADVANCED

A1 — Non-routine substitution
Solve $2(9^x) - 7(3^x) + 3 = 0$. Give all exact real solutions.

A2 — Parameter and number of log solutions
For $k > 0$, consider $\ln(x-1) + \ln(5-x) = \ln k$. Determine the number of distinct real solutions for (a) $0 < k < 4$, (b) $k = 4$, (c) $k > 4$. Explain the role of the logarithmic domain.

A3 — Exponential inequality by substitution (Phase 4 revised)
Solve $2(4^x) - 5(2^x) + 2 \leq 0$.

A4 — Continuous-model reconstruction
$P(t) = P_0 e^{kt}$ with $P(2) = 150$ and $P(5) = 240$. Find k exactly, P_0 exactly, numerical values, and the doubling time.

A5 — Exponential linearisation (Phase 4 revised)
$y = ka^x$. A graph of $\ln y$ against x passes through $(1, \ln 18)$ and $(3, \ln 162)$. Find the gradient, a, k, and the original relationship.

A6 — Reconstruct from the inverse
$f^{-1}(x) = 2\ln((x-1)/3)$ and $f(x) = Ae^{kx} + c$. Find A, k, c and the domain/range of both functions.

A7 — Reverse model comparison
$A(t) = 100b^t$, $B(t) = 400(0.8)^t$, and they are equal at $t = 4$. Find exact b, the percentage growth rate of A, which model is larger at $t = 0$, and interpret the intersection.

A8 — Transformed log and inverse
For $g(x) = 2\ln(3x-6) + 4$, find the domain, vertical asymptote, exact x-intercept, $g^{-1}(x)$, and inverse domain/range.

A9 — Parameterised exponential equation
For $e^{2x} - (k+1)e^x + k = 0$, determine all k giving (a) two distinct real solutions and (b) exactly one distinct real solution. Explain why positivity matters.

A10 — Power-law linearisation
$y = kx^n$ with $x, y > 0$. A $\ln y$ against $\ln x$ graph passes through $(0, \ln 3)$ and $(\ln 4, \ln 24)$. Find n, k, the original relationship, and y when $x = 9$.

WORKED SOLUTIONS — CORE

C1
A has constant difference 5, so it is linear. B has constant ratio $18/12 = 27/18 = 3/2$, so it is exponential.

C2
$A = 7$, $b = 3$, so $\boxed{f(x) = 7 \cdot 3^x}$. Then $f(2) = 63$. Since $3 > 1$, it is growth.

C3
$f(0) = 5$, so the intercept is $(0, 5)$. Domain $\mathbb{R}$; range $y > 0$; horizontal asymptote $y = 0$; increasing because $3/2 > 1$.

C4
Asymptote $y = -6$, range $y > -6$, $f(1) = 2 - 6 = -4$. Root: $2 \cdot 3^{x-1} = 6 \Rightarrow 3^{x-1} = 3 \Rightarrow \boxed{x = 2}$.

C5
$8 = 2^3$, $4 = 2^2$, so $2^{3x-3} = 2^{2x+2}$. Hence $3x - 3 = 2x + 2$ and $\boxed{x = 5}$.

C6
(a) $\boxed{\log_5 125 = 3}$. (b) $\boxed{2^{-3} = 1/8}$. (c) $\boxed{\log_7 1 = 0}$.

C7
$\log_2 40 - \log_2 5 + \log_2 2 = \log_2(40 \cdot 2/5) = \log_2 16 = \boxed{4}$.

C8
$f^{-1}(x) = \boxed{\log_4 x}$. Points swap to $(1/4, -1), (1, 0), (4, 1)$. Domain $x > 0$, range $\mathbb{R}$, vertical asymptote $x = 0$.

C9
Initial population 800; growth 5%. $P(3) = 800(1.05)^3 = 926.1 \approx \boxed{926}$. The factor 1.05 means each period the previous value is multiplied by 1.05.

C10
Domain $x - 2 > 0 \Rightarrow x > 2$. Then $x - 2 = 3^2 = 9$, so $\boxed{x = 11}$, which is valid.

WORKED SOLUTIONS — CHALLENGE

H1
$2^x = 17/5$, so $\boxed{x = \ln(17/5)/\ln 2} \approx \boxed{1.77}$.

H2
Let $u = 3^x > 0$. Then $u^2 - 10u + 9 = 0 = (u - 1)(u - 9)$, so $u = 1$ or 9. Hence $\boxed{x = 0, 2}$.

H3
Domain: $x > 3$. Combine logs: $\log_2[(x-1)(x-3)] = 3$, so $(x-1)(x-3) = 8$. Thus $x^2 - 4x - 5 = 0$, giving $x = 5, -1$. Only $\boxed{x = 5}$ is valid.

H4
Domain $2x - 4 > 0 \Rightarrow \boxed{x > 2}$; vertical asymptote $\boxed{x = 2}$. At $x = 5/2$, argument $= 1$, so $f = -1$. Root: $\ln(2x - 4) = 1/3$, hence $\boxed{x = 2 + \frac{1}{2}e^{1/3}}$.

H5
Domain $\mathbb{R}$, range $(-3, \infty)$. From $y = 4e^{x/2} - 3$, $e^{x/2} = (y + 3)/4$, so $\boxed{f^{-1}(x) = 2\ln((x + 3)/4)}$. Inverse domain $x > -3$, range $\mathbb{R}$.

H6
From $625 = 320b^3$, $b^3 = 625/320 = 125/64 = (5/4)^3$. Thus $\boxed{b = 5/4 = 1.25}$, growth rate $\boxed{25\%}$. $P(5) = 320(5/4)^5 = \boxed{976.5625}$.

H7
$300(1.04)^t = 250(1.08)^t \Rightarrow 1.2 = (1.08/1.04)^t$. Hence $\boxed{t = \ln 1.2/\ln(1.08/1.04)} \approx \boxed{4.83}$. At this time the models have equal value.

H8
$5(0.6)^x \leq 1 \Rightarrow (0.6)^x \leq 0.2$. Taking logs: $x \ln 0.6 \leq \ln 0.2$. Since $\ln 0.6 < 0$, divide and reverse: $\boxed{x \geq \ln 0.2/\ln 0.6} \approx \boxed{3.15}$.

H9
$f(0) = 4 \Rightarrow A + 1 = 4 \Rightarrow A = 3$. $f(2) = 13 \Rightarrow 3b^2 + 1 = 13 \Rightarrow b^2 = 4$. Since $b > 0$, $b = 2$. Thus $\boxed{f(x) = 3 \cdot 2^x + 1}$, asymptote $y = 1$, range $y > 1$.

H10
Let $u = \log_2 x$. Then $\log_4 x = u/2$. Hence $u + u/2 = 6$, so $u = 4$. Therefore $\boxed{x = 16}$.

WORKED SOLUTIONS — ADVANCED

A1
Let $u = 3^x > 0$. Then $2u^2 - 7u + 3 = 0 = (2u-1)(u-3)$, so $u = 1/2$ or 3. Hence $\boxed{x = -\ln 2/\ln 3}$ or $\boxed{x = 1}$.

A2
Domain: $1 < x < 5$. Combine: $(x-1)(5-x) = k = 4 - (x-3)^2$. Thus $0 < k < 4$ gives $\boxed{2}$ solutions, $k = 4$ gives $\boxed{1}$, and $k > 4$ gives $\boxed{0}$. The endpoints $x = 1, 5$ are excluded because the log arguments would be zero.

A3
Let $u = 2^x > 0$. Then $2u^2 - 5u + 2 \le 0 = (2u-1)(u-2) \le 0$, so $1/2 \le u \le 2$. Thus $2^{-1} \le 2^x \le 2^1$, hence $\boxed{-1 \le x \le 1}$.

A4
From $150 = P_0 e^{2k}$ and $240 = P_0 e^{5k}$, divide to get $8/5 = e^{3k}$. Hence $\boxed{k = \frac{1}{3}\ln(8/5)} \approx 0.1567$. Then $\boxed{P_0 = 150(8/5)^{-2/3}} \approx 109.7$. Doubling time $\boxed{t = 3\ln 2/\ln(8/5)} \approx 4.42$.

A5
$\ln y = \ln k + x \ln a$. Gradient

$$m = \frac{\ln 162 - \ln 18}{3 - 1} = \frac{\ln 9}{2} = \ln 3,$$

so $\boxed{a = 3}$. Using $(1, \ln 18)$: $\ln 18 = \ln k + \ln 3$, so $\ln k = \ln 6$ and $\boxed{k = 6}$. Hence $\boxed{y = 6 \cdot 3^x}$.

A6
Let $y = 2\ln((x-1)/3)$. Then $e^{y/2} = (x-1)/3$, so $x = 3e^{y/2} + 1$. Thus $\boxed{f(x) = 3e^{x/2} + 1}$, so $A = 3$, $k = 1/2$, $c = 1$. f: domain $\mathbb{R}$, range $(1, \infty)$. f^{-1}: domain $(1, \infty)$, range $\mathbb{R}$.

A7
$100b^4 = 400(0.8)^4 \Rightarrow b^4 = 4(0.8)^4$. Thus $\boxed{b = 0.8\sqrt{2} = 4\sqrt{2}/5} \approx 1.131$, so growth $\approx \boxed{13.1\%}$. At $t = 0$, $B(0) = 400 > A(0) = 100$. At $t = 4$ the graph intersection means the two quantities are equal.

A8
Domain $3x - 6 > 0 \Rightarrow \boxed{x > 2}$; vertical asymptote $x = 2$. Root: $2\ln(3x-6) + 4 = 0 \Rightarrow 3x - 6 = e^{-2}$, so $\boxed{x = 2 + e^{-2}/3}$. Inverse: $\boxed{g^{-1}(x) = 2 + \frac{1}{3}e^{(x-4)/2}}$. Its domain is $\mathbb{R}$, range $y > 2$.

A9
Let $u = e^x > 0$. Then $u^2 - (k+1)u + k = (u-1)(u-k) = 0$. $u = 1$ always gives $x = 0$. The second root is valid only if $k > 0$; it is distinct unless $k = 1$. Hence two distinct real solutions for $\boxed{k > 0,\ k \ne 1}$; exactly one for $\boxed{k \le 0 \text{ or } k = 1}$.

A10
$\ln y = \ln k + n \ln x$. Thus n is the gradient:

$$n = \frac{\ln 24 - \ln 3}{\ln 4} = \frac{\ln 8}{\ln 4} = \boxed{3/2}.$$

At $\ln x = 0$, intercept $= \ln 3$, so $\boxed{k = 3}$. Hence $\boxed{y = 3x^{3/2}}$. At $x = 9$, $y = 3 \cdot 27 = \boxed{81}$.

Hint ladders and safe randomization

Hints

Exponential equation: common base? If not, isolate then log.
Log equation: establish all argument restrictions first.
Substitution: look for b^x and $(b^x)^2$; enforce $u > 0$.
Inverse: isolate the exponential, then use ln.
Model: identify initial value and multiplicative factor; divide observations to eliminate P_0.
Inequality: check whether the base grows or decays; if dividing by $\ln b < 0$, reverse the sign.
Linearisation: compare the logged equation with $Y = mX + c$.

Generation locks

Generate exponential parameters first, then derive graph features. Generate log arguments before domain/asymptotes. For substitution use $u = b^x > 0$. Generate one coherent model first, then derive observations. For linearisation generate k, a first, then derive $\ln a$ and $\ln k$. Never generate model observations, inverse graphs, or target parameters independently.

Visual anti-leak and reliability lock

State 0 shows givens only; State 1 highlights structure; State 2 may reveal one conceptual relationship; State 3 may reveal one strong intermediate; State 4 shows the full solution. Do not pre-display target asymptotes, transformed solution graphs, target times, solved parameters, or accepted/rejected log candidates.

Every exponential/logarithmic graph must be generated from its actual equation. Inverse graphs must be produced by coordinate swapping/reflection across $y = x$, not by independent approximate drawing.

Canonical status

This Practice Bank contains the Phase 4 revised H6, revised A3 and revised A5, plus the final 10 Core + 10 Challenge + 10 Advanced question set with complete worked solutions.